

|                   |                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------|
| Experiment No.    | 1                                                                                                                   |
| Experiment Title  | Working with Python packages-Numpy, Scipy, Scikit-Learn, Matplotlib.                                                |
| GitHub Repository | <a href="https://github.com/Dharika-2006/ML-LAB-ASSIGNMENTS">https://github.com/Dharika-2006/ML-LAB-ASSIGNMENTS</a> |
| Student Name      | Dharika Shree K                                                                                                     |
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| Faculty           | Dr. Poreddy Ajay Kumar Reddy                                                                                        |
| Submission Date   | 02.08.26                                                                                                            |

## 1 Objective

To study the machine learning workflow by exploring Python libraries, performing EDA, preprocessing datasets, selecting relevant features, splitting the data into training, validation, and testing sets, evaluating model performance, and identifying the appropriate machine learning task and suitable algorithms for various datasets.

## 2 Problem Statement

Problem: To analyze machine learning datasets and perform the complete ML workflow using Python libraries.

Inputs:-

- i.) Loan amount prediction
- ii.) Handwritten character recognition
- iii.) Classification of Email spam and MNIST data
- iv.) Predicting Diabetes
- v.) Iris Dataset

Output: Preprocessed data, selected features, dataset splits, and evaluated machine learning model.

Objective: To understand and implement the machine learning workflow through EDA, preprocessing, feature selection, model building, and performance evaluation.

## 3 Dataset Description

|                    |                                           |
|--------------------|-------------------------------------------|
| Dataset Name       | Loan amount prediction                    |
| Dataset Source     | Kaggle                                    |
| Number of Samples  | 650                                       |
| Number of Features | 22                                        |
| Number of Classes  | 4                                         |
| Missing Values     | 137                                       |
| Train-Test Split   | 70% Training, 15% Validation, 15% Testing |

|                   |                                   |
|-------------------|-----------------------------------|
| Dataset Name      | Handwritten character recognition |
| Dataset Source    | Kaggle                            |
| Number of Samples | 1050                              |

|                    |                                           |
|--------------------|-------------------------------------------|
| Number of Features | 66                                        |
| Number of Classes  | 36                                        |
| Missing Values     | 43                                        |
| Train-Test Split   | 70% Training, 15% Validation, 15% Testing |

|                    |                                             |
|--------------------|---------------------------------------------|
| Dataset Name       | Classification of Email spam and MNIST data |
| Dataset Source     | Kaggle                                      |
| Number of Samples  | 1250                                        |
| Number of Features | 10                                          |
| Number of Classes  | 6                                           |
| Missing Values     | 83                                          |
| Train-Test Split   | 70% Training, 15% Validation, 15% Testing   |

|                    |                                           |
|--------------------|-------------------------------------------|
| Dataset Name       | Predicting Diabetes                       |
| Dataset Source     | Kaggle                                    |
| Number of Samples  | 1250                                      |
| Number of Features | 16                                        |
| Number of Classes  | 6                                         |
| Missing Values     | 179                                       |
| Train-Test Split   | 70% Training, 15% Validation, 15% Testing |

|                    |                                           |
|--------------------|-------------------------------------------|
| Dataset Name       | Iris Dataset                              |
| Dataset Source     | Kaggle                                    |
| Number of Samples  | 300                                       |
| Number of Features | 7                                         |
| Number of Classes  | 17                                        |
| Missing Values     | 10                                        |
| Train-Test Split   | 70% Training, 15% Validation, 15% Testing |

## 4 Exploratory Data Analysis

- **Loan Amount Prediction**

## Dataset overview :-

```
dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 650 entries, 0 to 649
Data columns (total 22 columns):
#   Column              Non-Null Count  Dtype
---  -
0   age                  477 non-null    float64
1   gender               532 non-null    object
2   marital_status       553 non-null    object
3   education_level      555 non-null    object
4   annual_income        470 non-null    float64
5   monthly_income      400 non-null    float64
6   employment_status    565 non-null    object
7   debt_to_income_ratio 424 non-null    float64
8   credit_score         488 non-null    float64
9   loan_amount          433 non-null    float64
10  loan_purpose           574 non-null    object
11  interest_rate        333 non-null    float64
12  loan_term            563 non-null    float64
13  installment          404 non-null    float64
14  grade_subgrade       612 non-null    object
15  num_of_open_accounts 424 non-null    float64
16  total_credit_limit    433 non-null    float64
17  current_balance       395 non-null    float64
18  delinquency_history   528 non-null    object
19  public_records        511 non-null    float64
20  num_of_delinquencies 561 non-null    float64
21  loan_paid_back        513 non-null    object
dtypes: float64(14), object(8)
memory usage: 111.8+ KB
```

Figure 1: Dataset Information Showing Feature Names, Data Types, and Non-Null Values

**Inference:** The dataset contains 650 records and 22 features, consisting of both numerical (float64) and categorical (object) attributes. The non-null count for each feature reveals that several columns contain missing values, indicating the need for data preprocessing. This overview provides a clear understanding of the dataset's structure and helps determine appropriate preprocessing and feature engineering steps before model training.

## Statistical summary

```
summary:
count    age    gender    marital_status    education_level    annual_income    \
count  477.000000    532    553    555    470.000000
unique    NaN    FEMALE    Single    Master    NaN
top    NaN    109    107    NaN
freq    44.838574    NaN    NaN    NaN    116871.638288
mean    26.887431    NaN    NaN    NaN    103838.316512
std    15.000000    NaN    NaN    NaN    5004.000000
min    17.000000    NaN    NaN    NaN    8778.750000
25%    42.000000    NaN    NaN    NaN    101185.000000
50%    74.000000    NaN    NaN    NaN    215400.500000
75%    85.000000    NaN    NaN    NaN    299631.000000
max

monthly_income    employment_status    debt_to_income_ratio    credit_score    \
count    400.000000    565    424.000000    488.000000
unique    NaN    6    NaN    NaN
top    NaN    Student    NaN    NaN
freq    NaN    98    NaN    NaN
mean    9258.576700    NaN    59.312099    582.827869
std    7758.915292    NaN    32.728207    272.992435
min    -1509.000000    NaN    5.350000    200.000000
25%    1589.400000    NaN    31.502500    276.000000
50%    8438.830000    NaN    58.140000    628.000000
75%    14760.000000    NaN    86.902500    867.000000
max    26198.500000    NaN    119.820000    900.000000

loan_amount    loan_term    installment    grade_subgrade    \
count    433.000000    563.000000    404.000000    612
unique    NaN    NaN    NaN    16
top    NaN    NaN    NaN    B2
freq    NaN    NaN    NaN    47
mean    109488.720411    48.447602    3120.271906    NaN
std    71618.230609    24.001157    3668.912446    NaN
min    1143.000000    12.000000    49.500000    NaN
25%    47076.000000    24.000000    1150.880000    NaN
50%    97704.000000    48.000000    2351.000000    NaN
75%    165153.000000    72.000000    3879.145000    NaN
max    249584.000000    84.000000    23910.090000    NaN

num_of_open_accounts    total_credit_limit    current_balance    \
count    424.000000    433.000000    395.000000
unique    NaN    NaN    NaN
top    NaN    NaN    NaN
freq    NaN    NaN    NaN
mean    15.294811    119181.011547    80238.762025
std    9.004090    69537.010685    72351.578274
min    0.000000    2191.000000    447.000000
25%    7.000000    59348.000000    26119.000000
50%    16.000000    116832.000000    47505.000000
75%    23.000000    172159.000000    123171.000000
max    30.000000    249465.000000    287043.000000

delinquency_history    public_records    num_of_delinquencies    \
count    528    511.000000    561.000000
unique    4    NaN    NaN
top    no    NaN    NaN
freq    144    NaN    NaN
mean    NaN    1.530333    2.388592
std    NaN    1.135246    1.686759
min    NaN    0.000000    0.000000
25%    NaN    1.000000    1.000000
50%    NaN    2.000000    2.000000
75%    NaN    3.000000    4.000000
max    NaN    3.000000    5.000000

loan_paid_back
count    513
unique    4
top    No
freq    133
mean    NaN
std    NaN
min    NaN
25%    NaN
50%    NaN
75%    NaN
max    NaN
```

Figure 2: Statistical Summary of the Loan Prediction Dataset



**Inference:** The target variable loanpaidback contains four encoded classes with a relatively similar number of samples in each category. The class distribution is fairly balanced, indicating that no single class dominates the dataset. A balanced target distribution helps reduce prediction bias and supports reliable classification model training.

## Correlation matrix

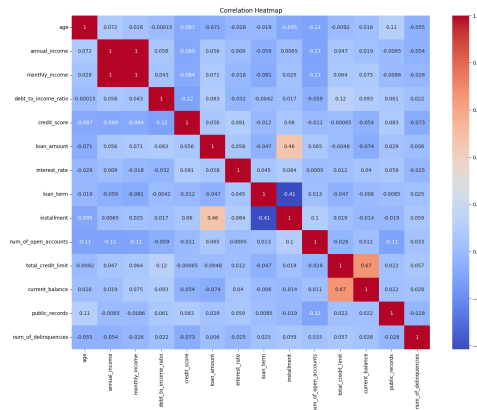


Figure 5: Correlation Heatmap of Numerical Features in the Loan Prediction Dataset

**Inference:** The heatmap illustrates the correlation coefficients between numerical features, helping identify relationships and potential feature dependencies. Most feature pairs exhibit weak correlations, indicating that the variables are largely independent and multicollinearity is minimal. A few feature pairs show moderate to strong correlations, such as annualincome–monthlyincome (1.00), totalcreditlimit–currentbalance (0.67), and loanamount–installment (0.46), which may be considered during feature selection and model building.

## Feature distribution

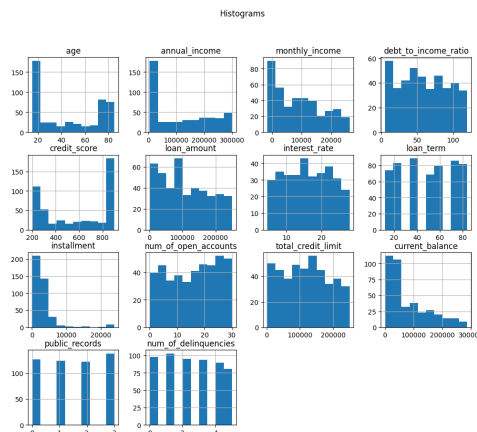


Figure 6: Histogram-Based Distribution of Numerical Features in the Loan Prediction Dataset

**Inference:** The histograms illustrate the distribution of values for each numerical feature, showing the frequency of observations across different value ranges. Several features, such as installment, currentbalance, and annualincome, exhibit right-skewed distributions, while others, such as interestrate and debttoincome ratio, appear more evenly distributed. Analyzing these distributions helps identify skewness, potential outliers, and data patterns, which are useful for selecting appropriate preprocessing techniques before model training.

## • Handwritten character recognition

Dataset overview :-

| dataset Info:                           |          |                |  |         |
|-----------------------------------------|----------|----------------|--|---------|
| <class 'pandas.core.frame.DataFrame'>   |          |                |  |         |
| *** RangeIndex: 1050 entries, 0 to 1049 |          |                |  |         |
| Data columns (total 66 columns):        |          |                |  |         |
| #                                       | Column   | Non-Null Count |  | Dtype   |
| 0                                       | pixel_0  | 1050 non-null  |  | float64 |
| 1                                       | pixel_1  | 1050 non-null  |  | float64 |
| 2                                       | pixel_2  | 1050 non-null  |  | float64 |
| 3                                       | pixel_3  | 1050 non-null  |  | float64 |
| 4                                       | pixel_4  | 1050 non-null  |  | float64 |
| 5                                       | pixel_5  | 1050 non-null  |  | float64 |
| 6                                       | pixel_6  | 1050 non-null  |  | float64 |
| 7                                       | pixel_7  | 1050 non-null  |  | float64 |
| 8                                       | pixel_8  | 1050 non-null  |  | float64 |
| 9                                       | pixel_9  | 1050 non-null  |  | float64 |
| 10                                      | pixel_10 | 1050 non-null  |  | float64 |
| 11                                      | pixel_11 | 1050 non-null  |  | float64 |
| 12                                      | pixel_12 | 1050 non-null  |  | float64 |
| 13                                      | pixel_13 | 1050 non-null  |  | float64 |
| 14                                      | pixel_14 | 1050 non-null  |  | float64 |
| 15                                      | pixel_15 | 1050 non-null  |  | float64 |
| 16                                      | pixel_16 | 1050 non-null  |  | float64 |
| 17                                      | pixel_17 | 1050 non-null  |  | float64 |
| 18                                      | pixel_18 | 1050 non-null  |  | float64 |
| 19                                      | pixel_19 | 1050 non-null  |  | float64 |
| 20                                      | pixel_20 | 1050 non-null  |  | float64 |
| 21                                      | pixel_21 | 1050 non-null  |  | float64 |
| 22                                      | pixel_22 | 1050 non-null  |  | float64 |
| 23                                      | pixel_23 | 1050 non-null  |  | float64 |
| 24                                      | pixel_24 | 1050 non-null  |  | float64 |
| 25                                      | pixel_25 | 1050 non-null  |  | float64 |
| 26                                      | pixel_26 | 1050 non-null  |  | float64 |
| 27                                      | pixel_27 | 1050 non-null  |  | float64 |
| 28                                      | pixel_28 | 1050 non-null  |  | float64 |
| 29                                      | pixel_29 | 1050 non-null  |  | float64 |
| 30                                      | pixel_30 | 1050 non-null  |  | float64 |
| 31                                      | pixel_31 | 1050 non-null  |  | float64 |
| 32                                      | pixel_32 | 1050 non-null  |  | float64 |
| 33                                      | pixel_33 | 1050 non-null  |  | float64 |
| 34                                      | pixel_34 | 1050 non-null  |  | float64 |
| 35                                      | pixel_35 | 1050 non-null  |  | float64 |
| 36                                      | pixel_36 | 1050 non-null  |  | float64 |
| 37                                      | pixel_37 | 1050 non-null  |  | float64 |
| 38                                      | pixel_38 | 1050 non-null  |  | float64 |
| 39                                      | pixel_39 | 1050 non-null  |  | float64 |
| 40                                      | pixel_40 | 1050 non-null  |  | float64 |
| 41                                      | pixel_41 | 1050 non-null  |  | float64 |
| 42                                      | pixel_42 | 1050 non-null  |  | float64 |
| 43                                      | pixel_43 | 1050 non-null  |  | float64 |
| 44                                      | pixel_44 | 1050 non-null  |  | float64 |
| 45                                      | pixel_45 | 1050 non-null  |  | float64 |
| 46                                      | pixel_46 | 1050 non-null  |  | float64 |
| 47                                      | pixel_47 | 1050 non-null  |  | float64 |
| 48                                      | pixel_48 | 1050 non-null  |  | float64 |
| 49                                      | pixel_49 | 1050 non-null  |  | float64 |
| 50                                      | pixel_50 | 1050 non-null  |  | float64 |
| 51                                      | pixel_51 | 1050 non-null  |  | float64 |
| 52                                      | pixel_52 | 1050 non-null  |  | float64 |
| 53                                      | pixel_53 | 1050 non-null  |  | float64 |
| 54                                      | pixel_54 | 1050 non-null  |  | float64 |
| 55                                      | pixel_55 | 1050 non-null  |  | float64 |
| 56                                      | pixel_56 | 1050 non-null  |  | float64 |
| 57                                      | pixel_57 | 1050 non-null  |  | float64 |
| 58                                      | pixel_58 | 1050 non-null  |  | float64 |
| 59                                      | pixel_59 | 1050 non-null  |  | float64 |
| 60                                      | pixel_60 | 1050 non-null  |  | float64 |
| 61                                      | pixel_61 | 1050 non-null  |  | float64 |
| 62                                      | pixel_62 | 1050 non-null  |  | float64 |
| 63                                      | pixel_63 | 1050 non-null  |  | float64 |
| 64                                      | label    | 1007 non-null  |  | float64 |
| 65                                      | issue    | 1050 non-null  |  | object  |
| dtypes: float64(65), object(1)          |          |                |  |         |
| memory usage: 541.5+ KB                 |          |                |  |         |

Figure 7: Dataset Information Showing Pixel Features, Target Label, and Data Types

**Inference:** The dataset contains 1,050 samples with 66 columns, comprising 65 numerical pixel features, 1 numerical label column, and 1 categorical issue column. All pixel features have 1,050 non-null values, indicating that there are no missing values in the dataset. The pixel values are stored as float64, making them suitable for numerical analysis and machine learning after appropriate preprocessing.

## Statistical summary

```

summary:
  pixel_0  pixel_1  pixel_2  pixel_3  pixel_4  \
count  1050.000000  1050.000000  1050.000000  1050.000000  1050.000000
unique    NaN      NaN      NaN      NaN      NaN
top      NaN      NaN      NaN      NaN      NaN
freq     NaN      NaN      NaN      NaN      NaN
mean    4.217543   8.775933   79.500190   170.579714   168.354686
std     17.350443   21.434923   74.928727   85.361935   85.091184
min      0.000000   0.000000   0.000000   0.000000   0.000000
25%      0.000000   0.000000   0.000000   127.500000   111.702500
50%      0.000000   0.000000   63.750000   195.500000   191.250000
75%      0.000000   1.447500   142.170000   239.000000   239.000000
max     255.000000  255.000000  255.000000  255.000000  255.000000

  pixel_5  pixel_6  pixel_7  pixel_8  pixel_9  ... \
count  1050.000000  1050.000000  1050.000000  1050.000000  1050.000000  ...
unique    NaN      NaN      NaN      NaN      NaN      ...
top      NaN      NaN      NaN      NaN      NaN      ...
freq     NaN      NaN      NaN      NaN      NaN      ...
mean    85.225810   20.366343   6.385952   5.844124   34.755924   ...
std     91.282067   48.142096   24.657310   22.732340   55.703806   ...
min      0.000000   0.000000   0.000000   0.000000   0.000000   ...
25%      0.000000   0.000000   0.000000   0.000000   0.000000   ...
50%      46.670000   0.000000   0.000000   0.000000   0.000000   ...
75%     159.380000   15.940000   0.000000   0.000000   47.810000   ...
max     255.000000  255.000000  255.000000  255.000000  255.000000   ...

  pixel_56  pixel_57  pixel_58  pixel_59  pixel_60  \
count  1050.000000  1050.000000  1050.000000  1050.000000  1050.000000
unique    NaN      NaN      NaN      NaN      NaN
top      NaN      NaN      NaN      NaN      NaN
freq     NaN      NaN      NaN      NaN      NaN
mean    6.361543   8.379029   85.92039   176.916048   169.021029
std     24.309233   23.935318   81.81478   86.270351   91.092333
min      0.000000   0.000000   0.000000   0.000000   0.000000
25%      0.000000   0.000000   0.000000   131.357500   111.617500
50%      0.000000   0.000000   63.750000   207.190000   206.520000
75%      0.000000   0.000000   145.48500   255.000000   255.000000
max      255.000000  255.000000  255.000000  255.000000  255.000000

  label  issue
count  1050  1050
unique    10  2
top      0    0
freq     43   1
mean    4.76  0.0095
std     2.05  0.083
min      0    0
25%      0    0
50%      0    0
75%      0    0
max      9    1

```

Figure 8: Statistical Summary of Pixel Features and Target Variables in the Handwritten Character Dataset

**Inference:** The statistical summary shows that the pixel intensity values range from 0 to 255, representing grayscale image data used for handwritten character recognition. Most pixel features exhibit high variability, indicating diverse handwriting patterns and stroke intensities across the dataset. The label column contains 1,007 valid entries with character classes ranging from 0 to 9, while the issue column identifies data quality information, with clean being the most frequent category.

## Missing value analysis

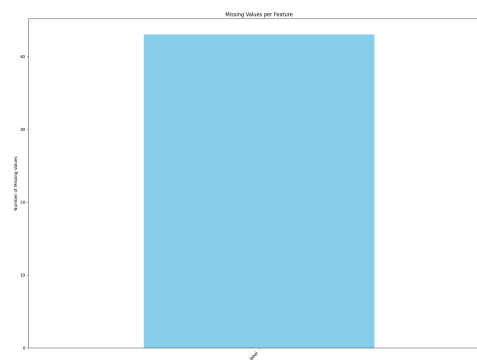


Figure 9: Missing Values Distribution Across Features in the Handwritten Character Dataset

**Inference:** The graph shows that the label column contains approximately 43 missing values, while all other features have no missing values. The pixel features are complete, ensuring that the input image data is available for all samples. The missing values in the label column should be handled through imputation or removal before training the classification model.

## Class distribution

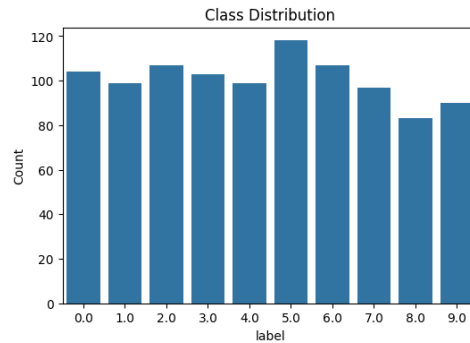


Figure 10: Class Distribution of Handwritten Character Labels

**Inference:** The dataset contains samples from 10 character classes (0–9), with each class represented by approximately 80–120 samples. Class 5 has the highest number of samples, while class 8 has the fewest, indicating a slight class imbalance. Overall, the class distribution is reasonably balanced, making the dataset suitable for training a classification model with minimal bias toward any particular class.

## Correlation matrix

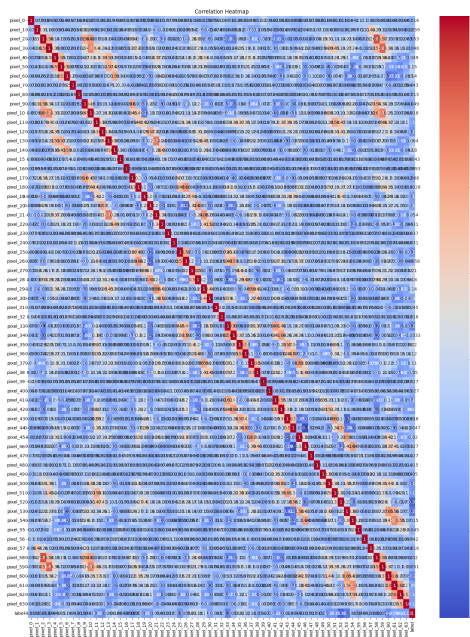


Figure 11: Correlation Heatmap of Pixel Features in the Handwritten Character Dataset

**Inference:** The heatmap illustrates the pairwise correlation among the 64 pixel features and the target label. Most pixel features exhibit weak to moderate correlations, indicating that each pixel captures distinct information about the handwritten characters. The label shows generally low correlation with individual pixel features, suggesting that character recognition depends on the combined pattern of multiple pixels rather than any single feature.

## Feature distribution

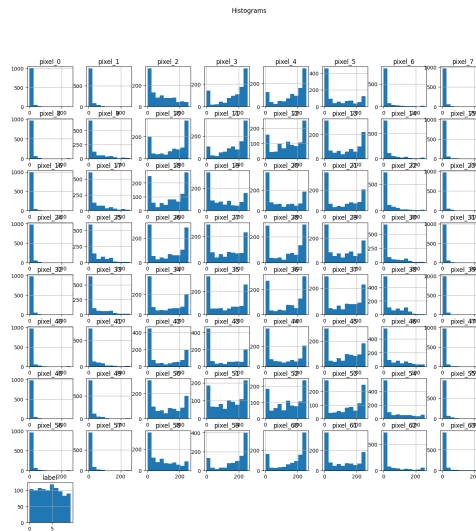


Figure 12: Histogram-Based Distribution of Pixel Intensity Features in the Handwritten Character Dataset

**Inference:** The histograms illustrate the distribution of pixel intensity values across all 64 pixel features in the dataset. Many pixel features are highly skewed toward lower intensity values, indicating that a large portion of the image pixels correspond to the background. Variations in the distributions of different pixel features reflect the diverse stroke patterns and shapes present in handwritten characters.

## • Classification of Email spam and MNIST data

### Dataset overview :-

```
dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1250 entries, 0 to 1249
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  -
0   email_id            915 non-null    object
1   subject             1100 non-null   object
2   message             1090 non-null   object
3   sender              1082 non-null   object
4   has_link            1001 non-null   object
5   num_links           1072 non-null   float64
6   num_attachments     1000 non-null   float64
7   message_length      730 non-null    float64
8   contains_special_chars 1004 non-null   object
9   label               1167 non-null   object
dtypes: float64(3), object(7)
memory usage: 97.8+ KB
```

Figure 13: Dataset Information Showing Email Spam Dataset Features, Data Types, and Non-Null Values

**Inference:** The dataset contains 1,250 email records with 10 features, including both numerical and categorical attributes relevant to spam classification. Several features, such as emailid, subject, message, sender, and messagelength, contain missing values and require preprocessing before model training. The target label

column has 1,167 non-null values, indicating that some records have missing class labels that should be handled before building the classification model.

## Statistical summary

```
summary:
  email_id subject message sender \
count 215 1180 1072 1082
unique 874 9 1072 1082
top 011 Hello Please send the report before noon! company.org 12
freq 1 120
mean NaN NaN NaN NaN
std NaN NaN NaN NaN
min NaN NaN NaN NaN
25% NaN NaN NaN NaN
50% NaN NaN NaN NaN
75% NaN NaN NaN NaN
max NaN NaN NaN NaN

  has_link num_links num_attachments message_length \
count 104 1072 1072 78
unique 1 1 1 78
top 0 NaN NaN NaN
freq 207
mean NaN 1.617472 0.403880 34.765479
std NaN 1.765426 1.111111 9.211000
min NaN 1.000000 1.000000 11.000000
25% NaN 6.000000 1.000000 22.000000
50% NaN 1.000000 1.000000 24.000000
75% NaN 5.000000 2.000000 35.000000
max NaN 5.000000 2.000000 67.000000

  contains_special_chars label \
count 104 1187
unique 1 1
top 0 0
freq 222 0
mean NaN NaN
std NaN NaN
min NaN NaN
25% NaN NaN
50% NaN NaN
75% NaN NaN
max NaN NaN

[ 'email_id', 'subject', 'message', 'sender', 'has_link', 'num_links', 'num_attachments', 'message_length', 'contains_special_chars', 'label' ]
Type: Tabular Dataset
```

Figure 14: Statistical Summary of the Email Spam Classification Dataset

**Inference:** The statistical summary presents descriptive statistics for both categorical (e.g., subject, message, sender) and numerical features (e.g., numlinks, numattachments, messagelength). The varying count values across features indicate the presence of missing values, which require preprocessing before model training. The dataset contains diverse email characteristics, such as message length, number of links, attachments, and special characters, which can be used as predictive features for spam classification.

## Missing value analysis

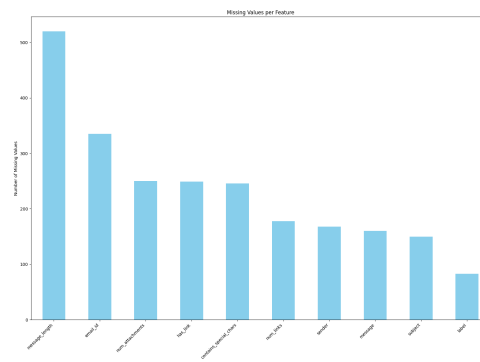


Figure 15: Missing Values Distribution Across Features in the Email Spam Classification Dataset

**Inference:** The graph shows that several features contain missing values, with messagelength having the highest number of missing entries, followed by emailid. Other features such as numattachments, haslink, containsspecialchars, numlinks, sender, message, subject, and label also contain missing values that require preprocessing. Proper handling of these missing values through imputation or removal is necessary to improve data quality and ensure reliable spam classification.

## Class distribution

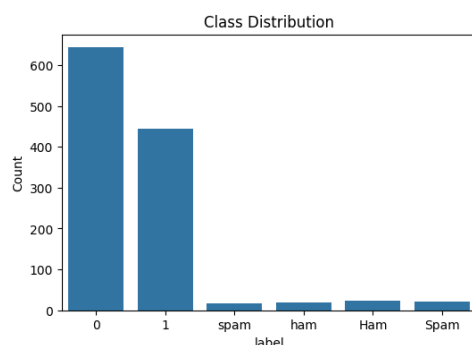


Figure 16: Class Distribution of Email Spam Labels

**Inference:** The dataset contains multiple representations of the target classes, including 0, 1, spam, ham, Ham, and Spam. The numeric labels (0 and 1) constitute the majority of the samples, while the text-based labels occur less frequently. The presence of inconsistent class labels indicates that the target variable should be standardized (e.g., converting all labels to Spam and Ham or 0 and 1) before training the classification model.

## Correlation matrix

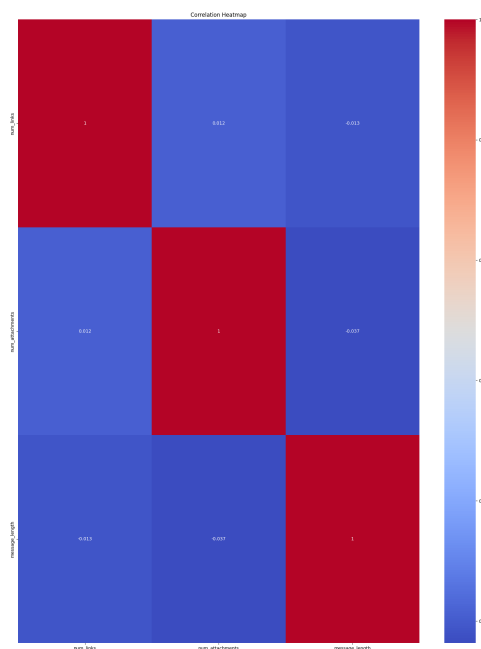


Figure 17: Correlation Heatmap of Numerical Features in the Email Spam Classification Dataset

**Inference:** The heatmap shows the pairwise correlations among the numerical features numlinks, numattachments, and messagelength. All feature pairs exhibit very weak correlations (correlation coefficients close to 0), indicating that these

features are largely independent of one another. The absence of strong correlations suggests that each numerical feature contributes distinct information for email spam classification.

## Feature distribution

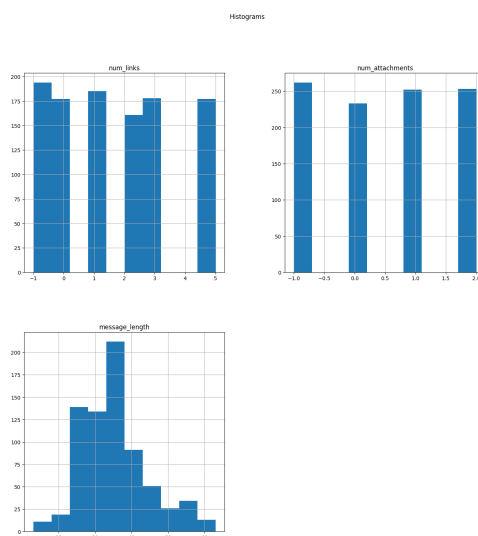


Figure 18: Histogram-Based Distribution of Numerical Features in the Email Spam Classification Dataset

**Inference:** The histograms illustrate the distributions of the numerical features `numlinks`, `numattachments`, and `messagelength`. `messagelength` exhibits a broader spread with most emails containing approximately 25–40 characters, while `numlinks` and `numattachments` are concentrated around a few discrete values. The varying distributions indicate that these numerical features capture different characteristics of emails and can contribute useful information for spam classification.

## • Predicting Diabetes

## Dataset overview :-

```
dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1250 entries, 0 to 1249
Data columns (total 16 columns):
#   Column              Non-Null Count  Dtype
---  -
0   patient_id          971 non-null    object
1   age                 926 non-null    float64
2   gender              1044 non-null   object
3   bmi                 832 non-null    float64
4   glucose_level       923 non-null    float64
5   blood_pressure      928 non-null    float64
6   insulin             884 non-null    float64
7   skin_thickness      844 non-null    float64
8   hba1c               645 non-null    float64
9   physical_activity    1042 non-null   object
10  family_history       1017 non-null   object
11  smoking_status       1041 non-null   object
12  cholesterol          882 non-null    float64
13  diabetes_pedigree_function 845 non-null    float64
14  pregnancies          829 non-null    float64
15  diabetes             1071 non-null   object
dtypes: float64(10), object(6)
memory usage: 156.4+ KB
```

Figure 19: Dataset Information Showing Feature Names, Data Types, and Non-Null Values of the Diabetes Prediction Dataset

**Inference:** The dataset contains 1,250 records with 16 features, comprising 10 numerical and 6 categorical attributes related to patient health and diabetes risk. Several features contain missing values, with hba1c (645 non-null) having the highest number of missing entries, followed by bmi, pregnancies, and cholesterol, indicating that preprocessing is required. The target variable diabetes has 1,071 non-null values and is categorical (object), making the dataset suitable for a supervised classification task after handling missing values and encoding categorical features.

## Statistical summary

```
summary:
patient_id  age  gender  bmi  glucose_level \
count      971  926.000000  1044  832.000000  923.000000
unique      934      NaN      5      NaN      NaN
top         345      NaN  Other      NaN      NaN
freq        2      NaN  227      NaN      NaN
mean      NaN  51.185745      NaN  42.405048  145.553629
std      NaN  34.010238      NaN  14.716605  103.140031
min      NaN   5.000000      NaN  15.100000  20.000000
25%      NaN  15.000000      NaN  29.975000  49.000000
50%      NaN  52.000000      NaN  44.900000  124.000000
75%      NaN  85.750000      NaN  55.050000  228.500000
max      NaN 100.000000      NaN  64.900000  350.000000

blood_pressure  insulin  skin_thickness  hba1c \
count      928.000000  884.000000  844.000000  645.000000
unique      NaN      NaN      NaN      NaN
top      NaN      NaN      NaN      NaN
freq      NaN      NaN      NaN      NaN
mean  111.734914  73.315611  16.015483  8.979380
std   57.162827  97.243893  19.832657  2.896516
min   30.000000  -5.000000  -1.000000  4.000000
25%   59.000000  -5.000000  -1.000000  6.400000
50%  107.000000  -5.000000  7.000000  8.500000
75%  161.000000  140.500000  31.000000  11.500000
max  220.000000  299.000000  60.000000  14.000000

physical_activity  family_history  smoking_status  cholesterol \
count      1042      1017      1041  882.000000
unique      5      4      5      NaN
top    High      no  CURRENT      NaN
freq      229      280      247      NaN
mean      NaN      NaN      NaN  143.557823
std      NaN      NaN      NaN  77.527759
min      NaN      NaN      NaN  50.000000
25%      NaN      NaN      NaN  82.000000
50%      NaN      NaN      NaN  113.000000
75%      NaN      NaN      NaN  196.000000
max      NaN      NaN      NaN  320.000000
```

```
diabetes_pedigree_function  pregnancies  smoking_status  cholesterol  physical_activity  family_history  gender  age  bmi  glucose_level  blood_pressure  insulin  skin_thickness  hba1c  diabetes
```

Figure 20: Statistical Summary of the Diabetes Prediction Dataset

**Inference:** The statistical summary provides descriptive measures such as count, mean, standard deviation, minimum, maximum, and quartiles for numerical features, along with unique values and frequency for categorical features. The dataset contains a mix of continuous variables (e.g., age, BMI, glucose level, blood pressure, cholesterol) and categorical variables (e.g., gender, physical activity, family history,



six classes are relatively similar, with no single class overwhelmingly dominating the dataset. Before training a classification model, the class labels should be standardized (e.g., mapping 0/No/Non-Diabetic to one class and 1/Yes/Diabetic to the other) to ensure a consistent binary target variable.

## Correlation matrix

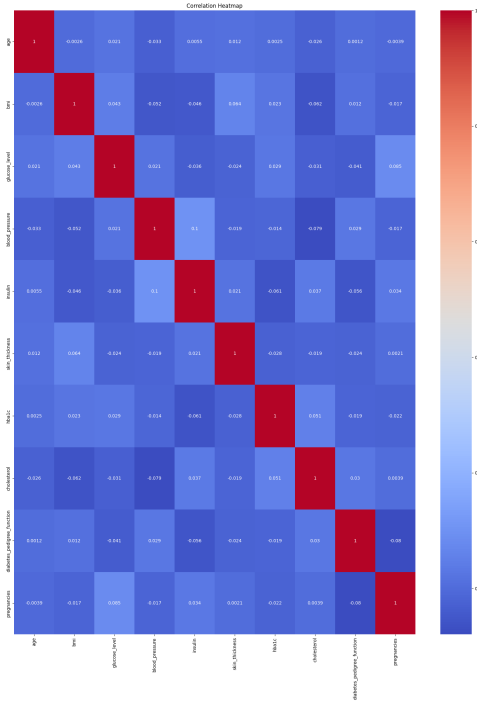


Figure 23: Correlation Heatmap of Numerical Features in the Diabetes Prediction Dataset

**Inference:** The heatmap shows the pairwise correlations among the numerical features, with diagonal values of 1.0 representing each feature's correlation with itself. Most correlation coefficients are close to zero (approximately -0.1 to 0.1), indicating weak linear relationships between the features and low multicollinearity. The absence of strong positive or negative correlations suggests that the numerical features provide largely independent information, making them useful for diabetes prediction without significant redundancy.

## Feature distribution

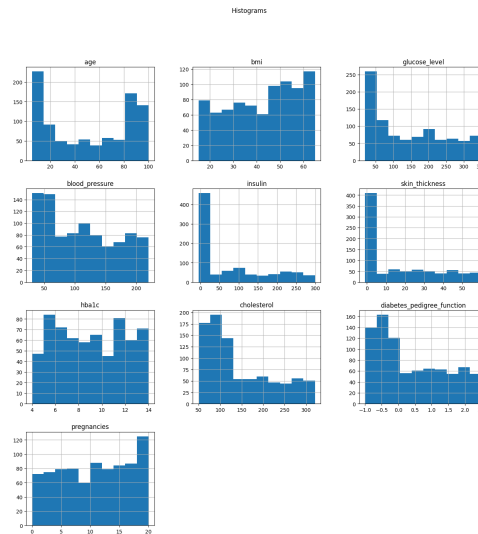


Figure 24: Histogram-Based Distribution of Numerical Features in the Diabetes Prediction Dataset

**Inference:** The histograms illustrate the distribution of the numerical features, including age, BMI, glucose level, blood pressure, insulin, skin thickness, HbA1c, cholesterol, diabetes pedigree function, and pregnancies. Several features, such as insulin, skin thickness, and glucose level, exhibit skewed distributions, indicating that patient measurements are not uniformly distributed across the dataset. The variation in feature distributions suggests that the dataset captures a wide range of patient health characteristics, which can provide valuable information for diabetes prediction.

## • Iris Dataset

### Dataset overview :-

```
dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   flower_id   300 non-null    object
1   sepal_length 280 non-null    float64
2   sepal_width  280 non-null    float64
3   petal_length 280 non-null    float64
4   petal_width  280 non-null    float64
5   species     290 non-null    object
6   Unnamed: 0   191 non-null    float64
dtypes: float64(5), object(2)
memory usage: 16.5+ KB
```

Figure 25: Dataset Information Showing Feature Names, Data Types, and Non-Null Values of the Iris Flower Dataset

**Inference:** The dataset contains 300 records with 7 features, including 5 numerical features and 2 categorical features. Several attributes, such as sepal\_length,

sepal\_width, petal\_length, and petal\_width, contain missing values (280 non-null), while Unnamed: 0 has the highest number of missing values (191 non-null). The target variable species has 290 non-null values and is categorical (object), indicating that the dataset is suitable for a supervised classification task after handling missing values and encoding categorical features.

## Statistical summary

```
summary:
flower_id  sepal_length  sepal_width  petal_length  petal_width  \
count      290      280.000000      280.000000      280.000000      280.000000
unique      158      NaN      NaN      NaN      NaN
top         15      NaN      NaN      NaN      NaN
freq         6      NaN      NaN      NaN      NaN
mean      NaN      6.534575      3.873214      3.726643      1.839643
std      NaN      3.261157      0.439923      1.774713      0.946016
min      NaN      4.300000      2.000000      1.000000      -2.400000
25%      NaN      5.100000      2.500000      1.500000      0.200000
50%      NaN      5.750000      3.000000      4.200000      1.300000
75%      NaN      6.500000      3.400000      5.100000      1.800000
max      NaN      25.165075      4.400000      6.900000      2.500000

species  Unnamed: 0
count      290      191.000000
unique       17      NaN
top      setosa      NaN
freq       37      NaN
mean      NaN      1.400211
std      NaN      0.911442
min      NaN      1.000000
25%      NaN      1.000000
50%      NaN      1.000000
75%      NaN      2.000000
max      NaN      7.000000

[["flower_id", "sepal_length", "sepal_width", "petal_length", "petal_width", "species", "Unnamed: 0"]]
Type: Tabular Dataset
```

Figure 26: Statistical Summary of the Iris Flower Dataset

**Inference:** The statistical summary presents descriptive measures such as count, mean, standard deviation, minimum, maximum, and quartiles for the numerical features, along with unique values and frequency for categorical features. The numerical features (sepal\_length, sepal\_width, petal\_length, and petal\_width) exhibit varying ranges and distributions, indicating differences in flower measurements across the dataset. The species column contains 17 unique categories, with setosa being the most frequent class, while the reduced counts of several features indicate the presence of missing values that should be handled before model training.

## Missing value analysis

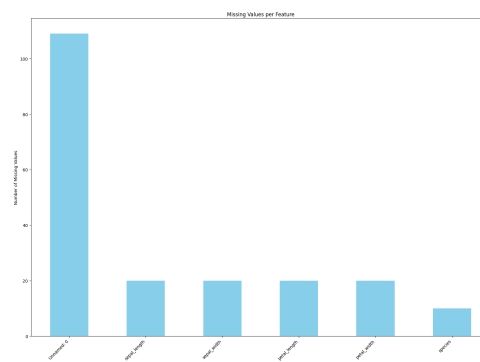


Figure 27: Missing Values Distribution Across Features in the Iris Flower Dataset

**Inference:** Missing values are present in six features, while flower\_id contains no missing values. The Unnamed: 0 column has the highest number of missing values (approximately 109), whereas sepal\_length, sepal\_width, petal\_length,

and petal\_width each contain about 20 missing values. The species target column has only 10 missing values, while the remaining missing values should be handled through appropriate imputation before model training.

## Class distribution

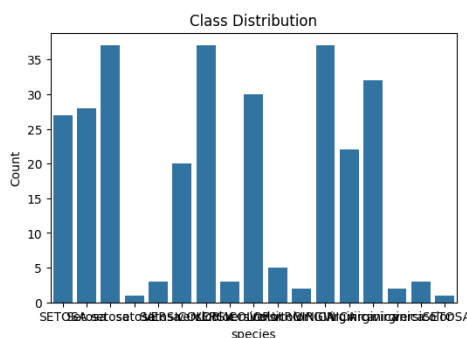


Figure 28: Class Distribution of the Iris Flower Dataset

**Inference:** The target variable species contains 17 distinct class labels, indicating inconsistent naming and representation of flower species. The class frequencies vary across categories, with some labels occurring much more frequently than others, resulting in an imbalanced distribution. Before training a classification model, the class labels should be standardized by merging different representations of the same species (e.g., differences in capitalization or spelling) to obtain consistent target classes.

## Correlation matrix

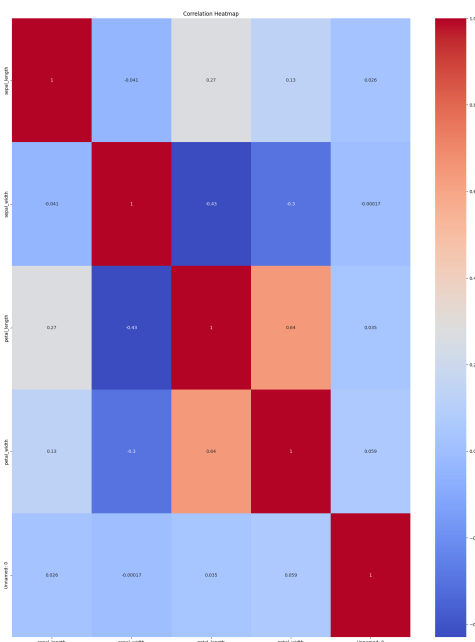


Figure 29: Correlation Heatmap of Numerical Features in the Iris Flower Dataset

**Inference:** The heatmap shows the pairwise correlations among the numerical features, with diagonal values of 1.0 representing each feature's correlation with itself. `petal_length` and `petal_width` exhibit the strongest positive correlation (0.64), indicating that flowers with longer petals tend to have wider petals. Most other feature pairs show weak to moderate correlations, suggesting that the numerical features capture different aspects of the iris flowers with limited redundancy.

## Feature distribution

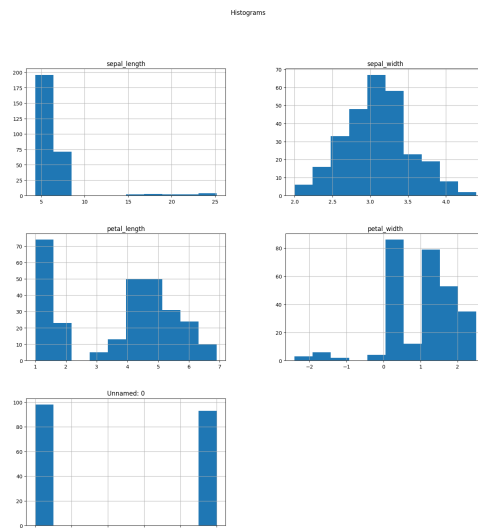


Figure 30: Histogram-Based Distribution of Numerical Features in the Iris Flower Dataset

**Inference:** The histograms illustrate the distributions of the numerical features, including sepal length, sepal width, petal length, petal width, and Unnamed: 0. The features exhibit different distribution patterns, with sepal width appearing relatively symmetric, while petal length and petal width show greater variation and multiple peaks across observations. The variation in feature distributions indicates that the flower measurements differ across samples, providing useful characteristics for distinguishing between iris species.

## 5 Data Preprocessing

### Loan Amount Prediction

- Missing value handling:
  - Missing values were identified using `df.isnull().sum()` and handled before model training.
  - Numerical features were imputed using their mean values, while categorical features were imputed using the mode (most frequent value).
  - This ensured that the dataset contained no missing values and prevented information loss caused by deleting records.

- Encoding
  - Categorical features such as gender, maritalstatus, educationlevel, employmentstatus, loanpurpose, gradesubgrade, and loanpaidback were converted into numerical form using Label Encoding.
  - Encoding transformed categorical values into machine-readable numerical labels while preserving all records for model training.
- Feature scaling
  - Numerical features, including age, annualincome, monthlyincome, loanamount, creditscore, interestrate, totalcreditlimit, and currentbalance, were standardized using StandardScaler.
  - Feature scaling ensures that all numerical variables have a similar scale, preventing features with larger values from dominating the learning process.
  - The target variable (loanpaidback) was excluded from scaling.
- Train-test split
  - The preprocessed dataset was divided into 70% training, 15% validation, and 15% testing sets.
  - Stratified sampling was applied when possible to preserve the class distribution across the subsets. If stratification was not feasible due to insufficient samples in a class, a standard random split was performed.

## Handwritten character recognition

- Missing value handling:
  - missing values were identified using `df.isnull().sum()` before preprocessing.
  - Only the label column contained missing values (43), while all 64 pixel features and the issue column had no missing values.
  - Records with missing label values were removed, as the target variable is essential for supervised classification.
- Encoding
  - The issue column, being categorical, was encoded into numerical values using Label Encoding to make it suitable for machine learning algorithms.
  - The label column already contained numerical class labels (0–9) and therefore did not require additional encoding.
  - The pixel features were numerical and required no categorical encoding.
- Feature scaling
  - The numerical pixel features were standardized using StandardScaler to ensure a uniform scale across all features.
  - Feature scaling improves the performance of machine learning algorithms by preventing features with larger values from dominating the learning process.

- The target (label) column was excluded from the scaling process.
- Train-test split
  - The dataset was divided into 618 training samples (70%), 133 validation samples (15%), and 133 testing samples (15%) to ensure reliable model development and evaluation.
  - Each dataset contains 65 input features, while the label column serves as the target variable for supervised classification.
  - The split enables model training on the training set, hyperparameter validation on the validation set, and unbiased performance assessment on the testing set.

## Classification of Email spam and MNIST data

- Missing value handling:
  - Missing values in numerical features (num\_links, num\_attachments, message\_length) were replaced using the mean of the respective feature.
  - Missing values in categorical features (email\_id, subject, message, sender, has\_link, contains\_special\_chars, label) were replaced using the mode (most frequent value).
  - This preprocessing step ensured that the dataset contained no missing values before feature selection and model training.
- Encoding
  - Categorical features such as email\_id, subject, message, sender, has\_link, contains\_special\_chars, and label were converted into numerical values using Label Encoding.
  - Label Encoding assigns a unique integer to each distinct category, making the data compatible with machine learning algorithms.
  - The numerical features (num\_links, num\_attachments, and message\_length) were already in numeric form and therefore required no encoding.
- Feature scaling
  - The numerical features (num\_links, num\_attachments, and message\_length) were standardized using StandardScaler.
  - StandardScaler transforms each numerical feature to have a mean of 0 and a standard deviation of 1, ensuring a common scale across all features.
  - Categorical features and the target label were excluded from the scaling process.
- Train-test split
  - The dataset was divided into Training (784 samples), Validation (168 samples), and Testing (169 samples) to support model training, hyperparameter tuning, and performance evaluation.

- Each sample contains 8 input features, resulting in feature dimensions of (784, 8) for training, (168, 8) for validation, and (169, 8) for testing.
- Since the label column is the target variable, the dataset was identified as a Supervised Learning Classification problem.

## Predicting Diabetes

- Missing value handling:
  - Missing values in numerical features (such as age, BMI, glucose\_level, blood\_pressure, insulin, skin\_thickness, hba1c, cholesterol, diabetes\_pedigree\_function, and pregnancies) were replaced using the mean of their respective columns.
  - Missing values in categorical features (patient\_id, gender, physical\_activity, family\_history, smoking\_status, and diabetes) were replaced using the mode (most frequent value).
  - This imputation process removed all missing values, resulting in a complete dataset suitable for preprocessing, feature selection, and model training.
- Encoding
  - Categorical features such as patient\_id, gender, physical\_activity, family\_history, smoking\_status, and the target variable diabetes were converted into numerical values using Label Encoding.
  - Label Encoding assigns a unique integer to each distinct category, making the categorical data suitable for machine learning algorithms.
  - The remaining features (age, BMI, glucose\_level, blood\_pressure, insulin, skin\_thickness, HbA1c, cholesterol, diabetes\_pedigree\_function, and pregnancies) were already numerical and therefore required no encoding.
- Feature scaling
  - All numerical features (age, BMI, glucose\_level, blood\_pressure, insulin, skin\_thickness, HbA1c, cholesterol, diabetes\_pedigree\_function, and pregnancies) were standardized using StandardScaler.
  - StandardScaler transforms each numerical feature to have a mean of 0 and a standard deviation of 1, ensuring that features with different ranges are placed on a common scale.
  - Categorical features (*patient\_id*, *gender*, *physical\_activity*, *family\_history*, *smoking\_status*) and
- Train-test split
  - The dataset was divided into Training (718 samples), Validation (154 samples), and Testing (154 samples) to enable model training, validation, and final performance evaluation.
  - Each dataset contains 14 input features, resulting in feature dimensions of (718, 14) for training, (154, 14) for validation, and (154, 14) for testing.
  - The diabetes column was selected as the target variable, identifying the problem as a Supervised Learning – Classification task.

## Iris Dataset

- Missing value handling:
  - Missing values in the numerical features (sepal\_length, sepal\_width, petal\_length, petal\_width, and Unnamed: 0) were replaced using the mean of their respective columns.
  - Missing values in the categorical feature (species) were replaced using the mode (most frequent class), while flower\_id required no imputation since it contained no missing values.
  - This imputation process eliminated all missing values, resulting in a complete dataset for subsequent preprocessing, feature selection, and classification.
- Encoding
  - The categorical features flower\_id and species were converted into numerical values using Label Encoding.
  - Label Encoding assigns a unique integer to each distinct category, enabling machine learning algorithms to process categorical data.
  - The remaining features (sepal\_length, sepal\_width, petal\_length, petal\_width, and Unnamed: 0) were already numerical and therefore required no encoding.
- Feature scaling
  - The numerical features (sepal\_length, sepal\_width, petal\_length, petal\_width, and Unnamed: 0) were standardized using StandardScaler.
  - StandardScaler transforms each numerical feature to have a mean of 0 and a standard deviation of 1, ensuring that all features are on a common scale.
  - The categorical features (flower\_id and species) were excluded from the scaling process after being encoded using Label Encoding.
- Train-test split
  - The dataset was divided into Training (193 samples), Validation (42 samples), and Testing (42 samples) for model training, validation, and performance evaluation.
  - Each sample consists of 4 input features, resulting in feature dimensions of (193, 4) for training, (42, 4) for validation, and (42, 4) for testing.
  - Since some species classes contained fewer than 2 samples, the dataset was split without stratification. The species column was selected as the target variable, identifying the problem as a Supervised Learning – Classification task.

## 6 Experimental Setup

---

|                      |              |
|----------------------|--------------|
| Python Version       | 3.14.0       |
| Scikit-Learn Version | 1.6.1        |
| IDE                  | Google Colab |
| Operating System     | Windows      |
| Hardware             | Laptop       |

---

## 7 Discussion

- Observation: Exploratory Data Analysis revealed variations in feature distributions, missing values, and class distributions across datasets, highlighting the importance of dataset-specific preprocessing before model development.
- Strength: A common machine learning pipeline was successfully applied to multiple datasets with minimal modifications.
- Limitation: Different dataset types (tabular, text, and image) require specialized preprocessing techniques, which are not fully handled by a single generic pipeline.
- Improvement: The pipeline can be enhanced by incorporating dataset-specific preprocessing methods, feature engineering, and hyperparameter optimization to improve overall performance.

## 8 Conclusion

The experiment successfully implemented a complete machine learning pipeline for multiple datasets, including dataset analysis, preprocessing, feature selection, data splitting, and model evaluation. Exploratory Data Analysis helped identify data characteristics such as missing values, feature distributions, and correlations, while preprocessing techniques prepared the data for machine learning. The study demonstrated that different datasets require different preprocessing strategies depending on their type (tabular, text, or image metadata), highlighting the importance of selecting appropriate techniques for accurate and reliable model development.

## 9 References

Datasets : <https://www.kaggle.com/>

IDE : <https://colab.research.google.com/>